

MANAS RAI

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EDUCATION

VIT Bhopal University

Bachelor of Technology in Computer Science Engineering

Bhopal, India

Oct 2023 – May 2027

TECHNICAL SKILLS

Programming Languages: C++, Java, Python, JavaScript

Web Technologies: HTML, CSS, React.js, Node.js

Developer Tools: Git, GitHub, VS Code

Core Competencies: Data Structures & Algorithms, Object-Oriented Programming, Problem Solving, System Design

Soft Skills: Public Relations, Team Collaboration, Quick Learning, Communication

PROJECTS

Inkwell – Multi-User Blogging Platform | React, Express.js, SQLite, JWT, Vite

- Developed a full-stack blogging platform supporting user registration, authentication, article publishing, commenting, and post engagement features.
- Implemented secure authentication using JWT-based authorization and bcrypt password hashing to protect user credentials and manage sessions.
- Built RESTful APIs for content creation, retrieval, updates, and user interactions, enabling efficient communication between frontend and backend services.
- Implemented engagement features including comments, likes, and author-based content management to enhance user interaction on the platform.

FluxLimiter – Configurable Rate Limiting System | Node.js, Express.js, Redis, Chart.js

- Built a pluggable rate limiter with 4 algorithms (token bucket, fixed window, sliding window log/counter) via a shared interface and Strategy pattern, decoupled from Express using a thin middleware adapter.
- Extended to a distributed setup using Redis and atomic Lua scripts, preventing race conditions in concurrent rate-limit checks across multiple server instances.
- Implemented lazy token refill (no background timers) and benchmarked algorithm tradeoffs (memory vs. accuracy vs. burst tolerance) with simulated traffic patterns.
- Unit-tested with deterministic time simulation, avoiding flaky setTimeout-based tests.

EXPERIENCE

GirlScript Summer of Code (GSSoC) 2025

Open-Source Contributor

Remote

Oct 2025 – Jan 2026

- Contributed to open-source projects as part of GSSoC 2025, one of India's largest open-source programs.

VCab - VIT Transportation Service

Public Relations Team Member

Bhopal, India

Oct 2023 – Jan 2024

- Managed communication strategies and stakeholder engagement for VIT's official transportation service

ACHIEVEMENTS & CERTIFICATIONS

- Certifications: Frontend Development Libraries (freeCodeCamp), Machine Learning Specialization (NPTEL), Market Analytics (NPTEL), MATLAB Certified Associate (MathWorks), AWS Technical Essentials (AWS)
- Participated in Smart India Hackathon (SIH), MP Police Hackathon, and competitive coding on LeetCode (Contest rating of 1403 with 250+ problems solved), and CodeVita